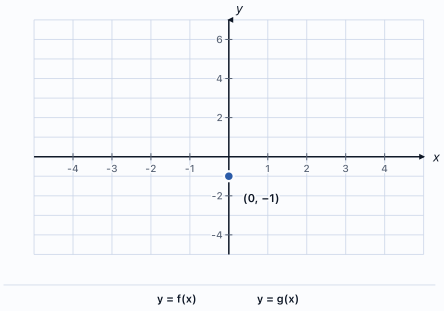
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Greens Theorem, Circulation & Planar Flux Practice Exercises

1 MCQ

Consider the function $z = f(x, y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.



A 10
B 13
C 26
D 52

2 MCQ

Evaluate the double integral of the function $f(x, y) = 2xy$ over the rectangular region $R = [0, 2] \times [0, 3]$ using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x.

A 12
B 18
C 6
D 24

3 MCQ

For $f(x, y) = 7x^2 + 4y^2$, find $f_y(-3, 1)$.

A -24
B 4
C 8
D -42

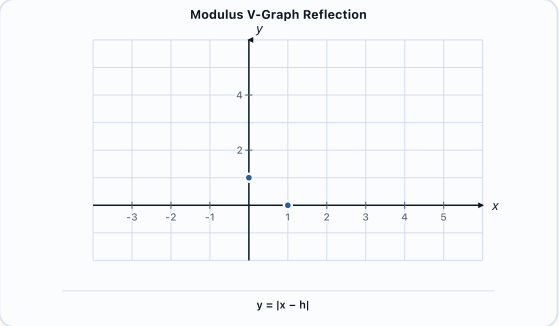
4 MCQ

Consider the function $f(x, y) = x^2 + y^2$. Find the value of $f_{xx}(2, 3)$.

A 2
B 4
C 6
D 0

5 MCQ

Let the linear scalar potential be $h(u, y) = 15u + 8y - 16$. Compute the squared gradient norm $\|\nabla h\|^2$.



A 289

B 293

C 281

D 304

6 MCQ

Consider the surface S which is the portion of the plane $z = 2x + 3y + 1$ lying above the rectangular region $R = [0, 1] \times [0, 2]$ in the xy-plane. Calculate the surface area of S.

A $\sqrt{14}$

B $2\sqrt{14}$

C $3\sqrt{14}$

D $4\sqrt{14}$

7 MCQ

Consider the vector field $\mathbf{u}(s, y, z) = \langle 22s,\ 7y,\ 7z \rangle$ on R^3. Compute the scalar divergence $\nabla \cdot \mathbf{u}$.

A 58

B 40

C 29

D 36

8 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. Evaluate the vector field at the point $(2, 4)$.

A $\langle 6, 8 \rangle$

B $\langle 8, 6 \rangle$

C $\langle 5, 6 \rangle$

D $\langle 6, 5 \rangle$

9 MCQ

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle -y, x, z \rangle$ around the circle $x^2 + y^2 = 9$ in the plane $z = 0$, oriented counterclockwise seen from above.

A 36π

B -18π

C 18π

D 9π

10 MCQ

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle y, -x, xz \rangle$ around the boundary of the square $[0,2] \times [0,2]$ in the plane $z = 0$, oriented counterclockwise seen from above.

A 8

B -4

C -8

D -16

11 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ across the sphere $x^2 + y^2 + z^2 = 4$.

A 64π

B 32π

C 16π

D -32π

12 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the cube $[0,2]^3$.

A 192

B 48

C -96

D 96

13 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle xz, 0, z^2 \rangle$ across the cube $[0,1]^3$.

A $\frac{3}{4}$

B $-\frac{3}{2}$

C 3

D $\frac{3}{2}$

14 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$ across the closed cylinder $x^2 + y^2 \le 1,\ 0 \le z \le 3$.

A 18π

B $\frac{9\pi}{2}$

C 9π

D -9π

15 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the sphere $x^2 + y^2 + z^2 = 4$.

A $-\frac{384\pi}{5}$

B $\frac{384\pi}{5}$

C $\frac{768\pi}{5}$

D $\frac{192\pi}{5}$

16 MCQ

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle y, x, 3 \rangle$ across the disk $x^2 + y^2 \leq 1,\ z = 0$, with upward normal.

A $\frac{3\pi}{2}$

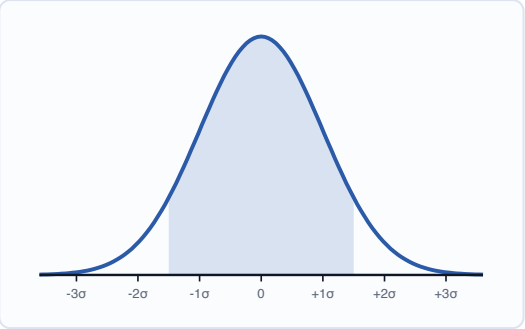
B 3π

C 6π

D -3π

17 MCQ

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle 0, 0, z \rangle$ across the part of the plane $x + y + z = 1$ in the first octant, with upward normal.



A $\frac{1}{6}$

B $\frac{1}{12}$

C $\frac{1}{3}$

D $-\frac{1}{6}$

18 MCQ

Calculate the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle 0, 0, z \rangle$ across the part of the plane $x + y + z = 3$ in the first octant, with upward normal.

A $\frac{9}{2}$

B 9

C $-\frac{9}{2}$

D $\frac{9}{4}$

19 MCQ

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle 1, 1, 1 \rangle$ across the part of the plane $x + y + z = 3$ in the first octant, with upward normal.

A $\frac{27}{2}$

B 27

C $-\frac{27}{2}$

D $\frac{27}{4}$

20 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle 2x, -y, 3z \rangle$ across the closed cylinder $x^2 + y^2 \leq 4,\ 0 \leq z \leq 2$.

A 64π

B 32π

C 16π

D -32π

21 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle -y, x \rangle$. Evaluate the line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where C is the circle $x^2 + y^2 = 4$ oriented counterclockwise, using Green's Theorem.

A 8π

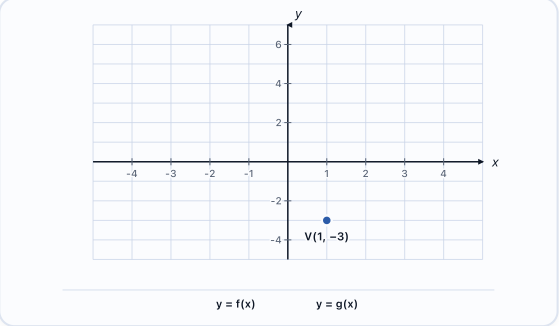
B 0

C 4π

D 16π

22 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$. According to Green's Theorem, the line integral of $\mathbf{F}$ around a simple closed curve C is equal to the double integral of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ over the region D enclosed by C, where $\mathbf{F} = \langle P, Q \rangle$. For the given vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$, identify the value of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$.



A x^2

B 0

C y

D -2x

23 MCQ

Calculate $\nabla f(-2, 2)$ for $f(x, y) = 2x^2 + 7y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	2	7	9

A (28, -8)

B (-8, 28)

C (-4, 14)

D (8, -28)

24 MCQ

Obtain $\nabla f(-5, -3)$ for $f(x, y) = 2x^2 + 3y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	2	3	5

A (-10, -9)

B (20, 18)

C (-18, -20)

D (-20, -18)

25 MCQ

Calculate $\nabla f(1, -6)$ for $f(x, y) = x^2 + 6y^2$.

Sample Surface Values f(x, y)

(x, y)	**(0, 0)**	**(1, 0)**	**(0, 1)**	**(1, 1)**
f(x, y)	0	1	6	7

A (-72, 2)

B (2, -72)

C (1, -36)

D (-2, 72)